

# NeuralGraph: complete results of the retrieval research campaign

September 5 to 6, 2026. All runs on LoCoMo (10 conversations, 1,540 questions in four categories: single\_hop 282, multi\_hop 841, temporal 321, open\_domain 96). Models: google/gemma-4-e4b for reranking, answering, and the campaign judge; nomic-embed-text v1.5 for embeddings; qwen/qwen3.6-35b-a3b as an independent judge. All numbers are leakage-free unless marked "old" (the December 2025 run used the gold answer as an acceptance gate and the gold category label for routing; both removed before any run below).

Two metrics are used throughout:

- **Accuracy**: an LLM judge labels the generated answer correct or wrong against the gold answer.
- **Recall@k**: any part of the gold answer (split on commas and "and", parts longer than two characters) appears as a substring in one of the top k retrieved messages. Retrieval-only, no LLM calls. Meaningless for the temporal category, whose gold answers are dates that live in metadata, so temporal is omitted from retrieval tables.

## 1. Headline: end-to-end accuracy of the shipped system

Shipped configuration: RETRIEVAL\_MODE=local\_pairs (per-agent memory routing, Tesseract top-50, 30 pair-node back-fills) plus open-domain prompt flags, 80-candidate pointwise rerank, 15 memories in the answer prompt (30 for list questions).

### 1.1 Conversations 1 to 5, 744 of 1,540 questions (run stopped early at the user's request)

| Category    | n   | New system (Gemma, lenient) | Old leaky run, same questions (GPT-4o, lenient) |
|-------------|-----|-----------------------------|-------------------------------------------------|
| single_hop  | 142 | 73.9%                       |                                                 |
| multi_hop   | 400 | 72.8%                       |                                                 |
| temporal    | 156 | 73.7%                       |                                                 |
| open_domain | 46  | 56.5%                       |                                                 |
| overall     | 744 | 72.2%                       | 66.9%                                           |

The two columns use different judges, so the 5.3-point gap is indicative only.

### 1.2 Conversations 1 to 4, 584 questions, under four judges

| Category    | n   | Old leaky (GPT-4o lenient) | Gemma lenient | Qwen lenient | Qwen strict | Substring |
|-------------|-----|----------------------------|---------------|--------------|-------------|-----------|
| single_hop  | 111 | 47.7                       | 75.7          | 62.2         | 33.3        | 21.6      |
| multi_hop   | 311 | 75.2                       | 72.3          | 66.6         | 41.8        | 27.0      |
| temporal    | 130 | 71.5                       | 78.5          | 71.5         | 60.8        | 36.2      |
| open_domain | 32  | 53.1                       | 62.5          | 43.8         | 6.2         | 3.1       |
| overall     | 584 | 68.0                       | 73.8          | 65.6         | 42.5        | 26.7      |

### 1.3 The clean comparison: single\_hop, original flat retrieval vs shipped stack

Same 282 questions, same code base, same judge, only retrieval changed.

| Retrieval                                                       | Accuracy     | recall@10 | recall@50 | seconds per question |
|-----------------------------------------------------------------|--------------|-----------|-----------|----------------------|
| flat (original Tesseract)                                       | 64.9%        | 39.4      | 61.7      | 6.4                  |
| hybrid (speaker boost + embedding back-fill)                    | 67.0%        | 44.7      | 61.7      | 8.2                  |
| graph (time chain, same-speaker edges, dialogue links appended) | 66.7%        | 39.4      | 61.7      | 7.9                  |
| <b>local_pairs (per-agent routing + pair back-fill)</b>         | <b>73.8%</b> | 46.8      | 62.8      | 7.9                  |

Same 111 single\_hop ids from conversations 1 to 4, flat vs shipped, per judge:

| Judge         | Flat | Shipped | Gain  | Wins / losses     |
|---------------|------|---------|-------|-------------------|
| Gemma lenient | 66.7 | 75.7    | +9.0  | 32 / 7, p < .001  |
| Qwen lenient  | 53.2 | 62.2    | +9.0  | 45 / 13, p < .001 |
| Qwen strict   | 22.5 | 33.3    | +10.8 | 38 / 11, p < .001 |
| Substring     | 18.0 | 21.6    | +3.6  | 18 / 5, p = .01   |

Where the flat system loses: when the gold text reached the final 15-memory context, accuracy was 77.8%; when it did not (124 of 282 questions), 48.4%.

## 2. Retrieval experiments (recall, no LLM calls)

### 2.1 Baseline retrievers, all 1,540 questions

| Retriever                                          | single_hop @10 / @50 | multi_hop @10 / @50 | open_domain @10 / @50 | ALL @10 / @50               |
|----------------------------------------------------|----------------------|---------------------|-----------------------|-----------------------------|
| pure embedding                                     | 25.9 / 52.1          | 35.1 / 54.8         | 13.5 / 19.8           | 25.3 / 41.9                 |
| Tesseract flat                                     | 39.4 / 61.7          | 46.8 / 61.4         | 13.5 / 20.8           | 34.9 / 48.0                 |
| Tesseract + graph neighbours mixed into top-50     | 27.3 / 38.3          | 38.3 / 43.5         | 7.3 / 10.4            | 27.5 / 32.9                 |
| Tesseract + graph neighbours appended (recall@all) | 39.4 / 61.7 / 64.5   | 46.8 / 61.4 / 62.2  | 13.5 / 20.8 / 21.9    | 34.9 / 48.0 / 49.0          |
| Tesseract + speaker boost                          | 44.7 / 61.7          | 49.2 / 61.4         | 13.5 / 20.8           | 37.2 / 48.0                 |
| Tesseract + embedding back-fill (recall@all)       | 39.4 / 61.7 / 65.2   | 46.8 / 61.4 / 65.6  | 13.5 / 20.8 / 22.9    | 34.9 / 48.0 / 65.5 (3 cats) |

Budget-matched conclusion: 30 extra candidates from graph neighbours add +1.4 recall@all; 30 from plain embedding add +4.1.

### 2.2 Pair nodes (H3): index each reply together with the message before it

| Variant                                   | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain        | ALL @10 / @50 / @all |
|-------------------------------------------|-----------------------------|----------------------------|--------------------|----------------------|
| pure embedding                            | 25.9 / 52.1 / 52.1          | 35.1 / 54.8 / 54.8         | 13.5 / 19.8 / 19.8 | 31.3 / 51.4 / 51.4   |
| embedding + pairs                         | 41.1 / 59.6 / 59.6          | 54.1 / 65.2 / 65.2         | 12.5 / 19.8 / 19.8 | 47.8 / 60.3 / 60.3   |
| Tesseract + embedding back-fill (control) | 39.4 / 61.7 / 65.2          | 46.8 / 61.4 / 65.6         | 13.5 / 20.8 / 22.9 | 42.5 / 58.2 / 62.2   |

| Variant                    | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain        | ALL @10 / @50 / @all |
|----------------------------|-----------------------------|----------------------------|--------------------|----------------------|
| Tesseract + pair back-fill | 39.4 / 61.7 / 68.4          | 46.8 / 61.4 / 69.1         | 13.5 / 20.8 / 24.0 | 42.5 / 58.2 / 65.4   |
| window of 3 messages       | 39.4 / 61.7 / 68.1          | 46.8 / 61.4 / 69.6         | 13.5 / 20.8 / 20.8 | 42.5 / 58.2 / 65.4   |

Verdict: works. +3.2 recall@all over the control at the same 30-candidate budget. Window of 3 adds nothing. Extra cost: 5,872 embeddings once.

### 2.3 Scoring-formula fixes (H4, H6)

| Configuration                                       | single_hop @10 / @50 | multi_hop @10 / @50 | open_domain @10 / @50 |
|-----------------------------------------------------|----------------------|---------------------|-----------------------|
| Tesseract baseline                                  | 39.4 / 61.7          | 46.8 / 61.4         | 13.5 / 20.8           |
| H4 keyword denominator fix (morphology + thesaurus) | 39.0 / 60.3          | 47.1 / 60.0         | 13.5 / 21.9           |
| H4 morphology only                                  | 39.0 / 61.3          | 47.9 / 60.6         | 11.5 / 20.8           |
| H6a soft temporal penalties                         | 39.4 / 61.7          | 46.8 / 61.5         | 13.5 / 20.8           |
| H6b no minimum-charge cutoff                        | 39.4 / 61.3          | 46.8 / 61.2         | 12.5 / 20.8           |
| H6c embedding rescue                                | 36.5 / 61.0          | 48.2 / 62.8         | 14.6 / 20.8           |
| H4 + H6 + rescue                                    | 36.5 / 59.9          | 47.9 / 62.0         | 13.5 / 20.8           |

Verdict: null. The four scoring stores overlap so heavily that per-store changes reorder nothing. Surprise: the temporal store dominates the fused top-10 for every category because fusion weights are near-uniform.

### 2.4 Two-agent memory split (H9a): treat each speaker as an agent with its own store

| Variant                                      | single_hop @10 / @50 | multi_hop @10 / @50 | open_domain @10 / @50 |
|----------------------------------------------|----------------------|---------------------|-----------------------|
| Tesseract, pooled (control)                  | 39.4 / 61.7          | 46.8 / 61.4         | 13.5 / 20.8           |
| Tesseract, routed to the named agent's store | 46.8 / 62.8          | 50.3 / 63.3         | 15.6 / 19.8           |
| Tesseract, oracle routing (gold speaker)     | 46.5 / 61.7          | 51.0 / 64.0         | 15.6 / 19.8           |
| Tesseract, federated by charge               | 36.9 / 60.3          | 42.4 / 59.9         | 11.5 / 20.8           |
| embedding, pooled                            | 25.9 / 52.1          | 35.1 / 54.8         | 13.5 / 19.8           |
| embedding, routed to the named agent         | 46.8 / 63.8          | 52.8 / 64.4         | 13.5 / 17.7           |

The gold-bearing message is spoken by the other agent only 2.9% of the time (0% for single\_hop), so name routing equals oracle routing. Federating by charge is worse than pooling because per-store charges are not comparable.

### 2.5 Speaker-swap probe (H9b): rewrite the question with the other speaker's name, 244 single\_hop questions

| Retriever      | Original recall@10 | Swapped recall@10 | Gap (identity sensitivity) | Still retrieves original gold in top 10 after swap |
|----------------|--------------------|-------------------|----------------------------|----------------------------------------------------|
| pure embedding | 26.6               | 34.8              | -8.2                       | 78.5%                                              |
| Tesseract      | 42.6               | 32.0              | +10.7                      | 57.7%                                              |

| Retriever                 | Original recall@10 | Swapped recall@10 | Gap (identity sensitivity) | Still retrieves original gold in top 10 after swap |
|---------------------------|--------------------|-------------------|----------------------------|----------------------------------------------------|
| Tesseract + speaker boost | 48.8               | 26.2              | +22.5                      | 45.4%                                              |

Swapping the name raises pure-embedding recall. Measured cause: a speaker's own name appears in 0.1% of their messages, the other speaker's name in 33.4%. A name in the question is an embedding-level pointer to the wrong speaker.

## 2.6 Combined stacks with local routing (H9 follow-up), 1,219 questions (three categories)

| Variant                                               | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain @10 / @50 / @all |
|-------------------------------------------------------|-----------------------------|----------------------------|------------------------------|
| embedding, local                                      | 46.8 / 63.8 / 63.8          | 52.8 / 64.4 / 64.4         | 13.5 / 17.7 / 17.7           |
| embedding + pairs, local                              | 41.1 / 59.2 / 59.2          | 53.7 / 64.0 / 64.0         | 14.6 / 20.8 / 20.8           |
| Tesseract, local                                      | 46.8 / 62.8 / 62.8          | 50.3 / 63.3 / 63.3         | 15.6 / 19.8 / 19.8           |
| <b>Tesseract local + pair back-fill (shipped)</b>     | 46.8 / 62.8 / <b>69.5</b>   | 50.3 / 63.3 / <b>68.4</b>  | 15.6 / 19.8 / <b>24.0</b>    |
| Tesseract local + plain embedding back-fill (control) | 46.8 / 62.8 / 66.0          | 50.3 / 63.3 / 65.4         | 15.6 / 19.8 / 19.8           |
| embedding local, top 80 (control)                     | 46.8 / 63.8 / 66.7          | 52.8 / 64.4 / 66.0         | 13.5 / 17.7 / 21.9           |

Pair nodes are a good back-fill and a bad base: inside the top-50 they cost single\_hop 5.7 points at recall@10 because expanding a pair pulls in the other agent's message.

## 2.7 Reply-only pair mapping (N3)

| Variant                               | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain @all |
|---------------------------------------|-----------------------------|----------------------------|------------------|
| shipped (both members)                | 46.8 / 62.8 / 69.5          | 50.3 / 63.3 / 68.4         | 24.0             |
| reply-only back-fill                  | 46.8 / 62.8 / 68.4          | 50.3 / 63.3 / 67.1         | 21.9             |
| embedding + pairs reply-only, as base | 45.7 / 62.1 / 62.1          | 55.4 / 64.2 / 64.2         | 18.8             |
| pairs only, reply-only                | 45.4 / 62.1 / 62.1          | 55.4 / 64.2 / 64.2         | 18.8             |

Verdict: negative. Of 23 questions lost, 14 had the gold only in the other agent's message; under local routing the both-member pair link is the only channel that lets it in.

## 2.8 Mega search (M1): vector + BM25 + entity-graph PageRank hop, fused by reciprocal rank, 1,219 questions

| Retriever                         | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain @10 / @50 / @all | ALL @10 / @50 / @all |
|-----------------------------------|-----------------------------|----------------------------|------------------------------|----------------------|
| shipped (Tesseract local + pairs) | 46.8 / 62.8 / 69.5          | 50.3 / 63.3 / 68.4         | 15.6 / 19.8 / 24.0           | 46.8 / 59.7 / 65.1   |
| vector + BM25                     | 46.8 / 62.1 / 65.2          | 55.2 / 66.0 / 67.4         | 12.5 / 18.8 / 19.8           | 49.9 / 61.4 / 63.2   |
| vector + graph hop                | 47.2 / 64.9 / 67.4          | 57.8 / 65.3 / 66.8         | 14.6 / 20.8 / 22.9           | 51.9 / 61.7 / 63.5   |
| vector + BM25 + graph             | 47.5 / 62.4 / 67.0          | 55.2 / 64.6 / 67.3         | 12.5 / 21.9 / 22.9           | 50.0 / 60.7 / 63.7   |

| Retriever                                               | single_hop @10 / @50 / @all | multi_hop @10 / @50 / @all | open_domain @10 / @50 / @all | ALL @10 / @50 / @all |
|---------------------------------------------------------|-----------------------------|----------------------------|------------------------------|----------------------|
| vector + graph + Tesseract, then pairs (regex entities) | 51.1 / 65.6 / 68.1          | 59.1 / 66.6 / 68.5         | 14.6 / 21.9 / 24.0           | 53.7 / 62.8 / 64.9   |
| same, LLM-built entity graph (M2)                       | 49.6 / 64.5 / 67.7          | 56.7 / 66.7 / 68.6         | 14.6 / 19.8 / 24.0           | 51.8 / 62.5 / 64.9   |
| same, LLM + regex entities                              | 50.7 / 64.9 / 68.4          | 58.9 / 66.7 / 68.5         | 14.6 / 20.8 / 24.0           | 53.5 / 62.7 / 65.0   |
| LLM graph channel alone                                 | 18.1 / 25.9 / 29.4          | 24.9 / 33.2 / 36.0         | 5.2 / 8.3 / 8.3              | 21.7 / 29.5 / 32.3   |

BM25 adds nothing once vectors and pairs exist. The graph hop is the only graph mechanism that beat plain embedding at the retrieval level (+6.9 recall@10 overall, +8.8 on multi\_hop). See section 3.7 for why it still loses end to end.

### 3. End-to-end experiments (accuracy, same 100 single\_hop ids unless noted, shipped retrieval, Gemma lenient judge)

Control for every row: 78 of 100. Gold text present in the final 15-memory context: 71%. Accuracy given gold present: 88.7%.

#### 3.1 Listwise reranker (H1)

| Reranker                                           | Accuracy | Rerank ms | End-to-end ms | Gold in context | Accuracy given gold | LLM calls per question |
|----------------------------------------------------|----------|-----------|---------------|-----------------|---------------------|------------------------|
| pointwise 0 to 3, one call per candidate (control) | 78       | 7,391     | 7,948         | 71%             | 88.7%               | 80                     |
| listwise, 8 batches of 10, 0 to 10                 | 74       | 3,258     | 3,816         | 71%             | 84.5%               | 8                      |
| listwise + second global pass                      | 72       | 5,845     | 6,387         | 71%             | 83.1%               | 9                      |

Verdict: latency win, accuracy loss. Gold-in-context is identical across all three: the reranker only reorders, retrieval decides.

#### 3.2 Answer context size (H2, N1)

| Memories in prompt (plain / list questions) | Accuracy  | Gold in context | Accuracy given gold | Answer ms |
|---------------------------------------------|-----------|-----------------|---------------------|-----------|
| 5 / 10                                      | 69        | 57%             | 87.7%               | 180       |
| 10 / 10                                     | 71        | 60%             | 88.3%               | 190       |
| 10 / 15                                     | 71        | 65%             | 83.1%               | 250       |
| <b>15 / 30 (default)</b>                    | <b>78</b> | 71%             | 88.7%               | 479       |
| 30 / 30                                     | 76        | 71%             | 87.3%               | 506       |
| 50 / 50                                     | 69        | 73%             | 78.1%               | 593       |

Post-rerank rank of the gold message: 1 to 5 for 52 questions, 6 to 15 for 15, 16 to 30 for 4, absent for 29. Widening adds no gold and distracts the model; narrowing loses corroborating second mentions. The default is the optimum.

### 3.3 LLM router vs regex router (H7)

| Category   | Regex (control) | LLM router | Agreement | Added latency |
|------------|-----------------|------------|-----------|---------------|
| single_hop | 78 / 100        | 79 / 100   | 52%       | +105 ms       |
| temporal   | 78 / 100        | 76 / 100   | 97%       | +119 ms       |

Verdict: null. Regex already sends 97 of 100 temporal questions to the temporal mode. The two routers disagree on half of single\_hop yet score the same.

### 3.4 Open-domain prompt flags (H5), all 96 open\_domain questions

| Configuration                                             | Accuracy        | Abstentions |
|-----------------------------------------------------------|-----------------|-------------|
| control                                                   | 41 / 96 = 42.7% | 27          |
| keep context in world mode + hedge routing                | 43 / 96 = 44.8% |             |
| + memories-plus-world-knowledge prompt for inference mode | 55 / 96 = 57.3% | 13          |

single\_hop with the flags on: 77 vs 78 (no regression). Under the strict independent judge the gain disappears (see 4.2); about 4 of the 17 wins are judge leniency on hedged answers.

### 3.5 Pair-aware reranker and answer prompt (N4)

| Configuration                        | Accuracy | Rerank ms | Gold in context | Accuracy given gold |
|--------------------------------------|----------|-----------|-----------------|---------------------|
| control                              | 78       | 7,349     | 71%             | 88.7%               |
| reranker sees previous message       | 71       | 8,479     | 69%             | 82.6%               |
| answer prompt shows previous message | 75       | 7,989     | 73%             | 83.6%               |
| both                                 | 66       | 8,197     | 70%             | 78.6%               |

Verdict: negative. The previous line dilutes the candidate rather than explaining it. Pair context belongs in retrieval only.

### 3.6 Answer model swap (H8): Gemma-4-e4b vs Qwen3.6-35b-a3b for the answer call only

|                                       | Gemma answers | Qwen answers |
|---------------------------------------|---------------|--------------|
| accuracy, Gemma lenient judge         | 78            | 71           |
| accuracy, Qwen lenient judge          | 64            | 65           |
| accuracy, Qwen strict judge           | 38            | 42           |
| substring                             | 24            | 29           |
| gold in context                       | 71            | 71           |
| "Not found" answers with gold present | 4             | 11           |
| answer latency ms                     | 501           | 913          |

Verdict: a judge artifact. The sign flips once the judge is not the answering model's family.

### 3.7 Mega search and LLM-built graph end to end (M1, M2)

| Category, same 100 ids | Shipped | Regex-entity mega search | LLM-entity mega search |
|------------------------|---------|--------------------------|------------------------|
| single_hop             | 78      | 68 (3 wins / 13 losses)  | 70 (4 / 12)            |
| multi_hop              | 75      | 76                       | 77 (8 / 6)             |

Accuracy given gold in context on single\_hop: 88.6 shipped, 79.1 regex graph, 82.6 LLM graph. The entity hop fills the candidate window with topically related non-answers that the reranker and answer model prefer. LLM extraction (138 to 354 resolved entities and 430 to 712 triples per conversation, versus about 2,000 regex entities) recovers 2 points and does not fix it.

## 4. Judge sensitivity (N5)

### 4.1 Same answer files, four graders

| Answer file               | n   | Gemma lenient (campaign) | Qwen lenient | Qwen strict | Substring |
|---------------------------|-----|--------------------------|--------------|-------------|-----------|
| flat, single_hop          | 282 | 64.9                     | 51.4         | 21.6        | 13.5      |
| shipped, single_hop       | 282 | 73.8                     | 62.8         | 31.2        | 18.1      |
| H8 Gemma answers          | 100 | 78.0                     | 64.0         | 38.0        | 24.0      |
| H8 Qwen answers           | 100 | 71.0                     | 65.0         | 42.0        | 29.0      |
| H5 control, open_domain   | 96  | 42.7                     | 27.1         | 11.5        | 6.2       |
| H5 treatment, open_domain | 96  | 57.3                     | 39.6         | 10.4        | 9.4       |

### 4.2 Does the verdict survive the judge?

| Pair                       | Gemma lenient | Qwen lenient | Qwen strict | Substring | Verdict                     |
|----------------------------|---------------|--------------|-------------|-----------|-----------------------------|
| flat to shipped            | +8.9          | +11.3        | +9.6        | +4.6      | stable, works under all     |
| Gemma to Qwen answers (H8) | -7.0          | +1.0         | +4.0        | +5.0      | sign flips, null            |
| open-domain flags (H5)     | +14.6         | +12.5        | -1.0        | +3.1      | vanishes under strict, null |

Agreement with the campaign judge: Qwen lenient 86.5% (kappa 0.72), Qwen strict 59.4% (kappa 0.30), substring 49.1% (kappa 0.17). Gemma-correct answers average 53 characters, wrong ones 21; the strict judge shows no length bias. Judge model alone moves scores about 12 points with an identical prompt; prompt leniency about 30 points, almost all on multi-item gold answers (66% of single\_hop golds).

## 5. Summary table of every experiment

| #      | Experiment                          | Level           | Result                              | Verdict                            |
|--------|-------------------------------------|-----------------|-------------------------------------|------------------------------------|
| 0      | Leakage removal                     | harness         | scores now honest                   | required                           |
| H9     | Per-agent memory routing            | retrieval + e2e | recall@10 +7.4; e2e +8.9 with pairs | <b>works</b>                       |
| H3     | Pair nodes as back-fill             | retrieval       | +3.2 recall@all                     | <b>works</b>                       |
| H5     | Open-domain prompt flags            | e2e             | +14.6 lenient, -1.0 strict          | lenient-judge only                 |
| hybrid | Speaker boost + embedding back-fill | e2e             | +2.1                                | small                              |
| graph  | Graph-neighbour expansion           | retrieval + e2e | +1.4 recall, +1.8 e2e               | no better than embedding back-fill |
| H1     | Listwise reranker                   | e2e             | -4, latency -56%                    | speed only                         |
| H2     | Wider context (30, 50)              | e2e             | -2, -9                              | hurts                              |

| #  | Experiment                              | Level           | Result                                           | Verdict        |
|----|-----------------------------------------|-----------------|--------------------------------------------------|----------------|
| N1 | Narrower context (10, 5)                | e2e             | -7, -9                                           | hurts          |
| H7 | LLM router                              | e2e             | +1 / -2                                          | null           |
| H4 | Keyword formula fix                     | retrieval       | -0.4 to -1.4                                     | null           |
| H6 | Softer temporal penalties               | retrieval       | within 0.2                                       | null           |
| N3 | Reply-only pairs                        | retrieval       | -1 to -2                                         | hurts          |
| N4 | Pair-aware reranker / prompt            | e2e             | -7 / -3 / -12                                    | hurts          |
| H8 | 35B answer model                        | e2e             | -7 lenient, +4 strict                            | judge artifact |
| M1 | Mega search (vector + BM25 + graph hop) | retrieval + e2e | +6.9 recall@10; e2e -10 single_hop, +1 multi_hop | hurts          |
| M2 | LLM-built entity graph in mega search   | retrieval + e2e | +5.0 recall@10; e2e -8 single_hop, +2 multi_hop  | hurts          |
| N5 | Judge sensitivity                       | evaluation      | 51-point spread by grader                        | paper-grade    |

Files: per-question results in `demo/results/*.json`; per-experiment reports in `docs/research/`; rescoring script `docs/research/n5_rescore.py`.